卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0313 30-B003701S-T0313

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 03 卷 第 13 册
Structure SPECIALITY VOLUME BOOK

卷册名称 直流场设备支架基础图（围栏内）（第一次修改）
Volume name foundations for ±660kV DC equipment support (in the fence)(first modification)

批 准 APPROVED BY	张孟忠
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	谢财齐
图 纸 6 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0313-001	直流场设备支架基础布置图（围栏内） The layout of foundations for ±660kV DC equipment support (in the fence)	B	1		
2	30-B003701S-T0313-002	设备支架基础汇总表 The summary table of foundations of equipment support	A	1		
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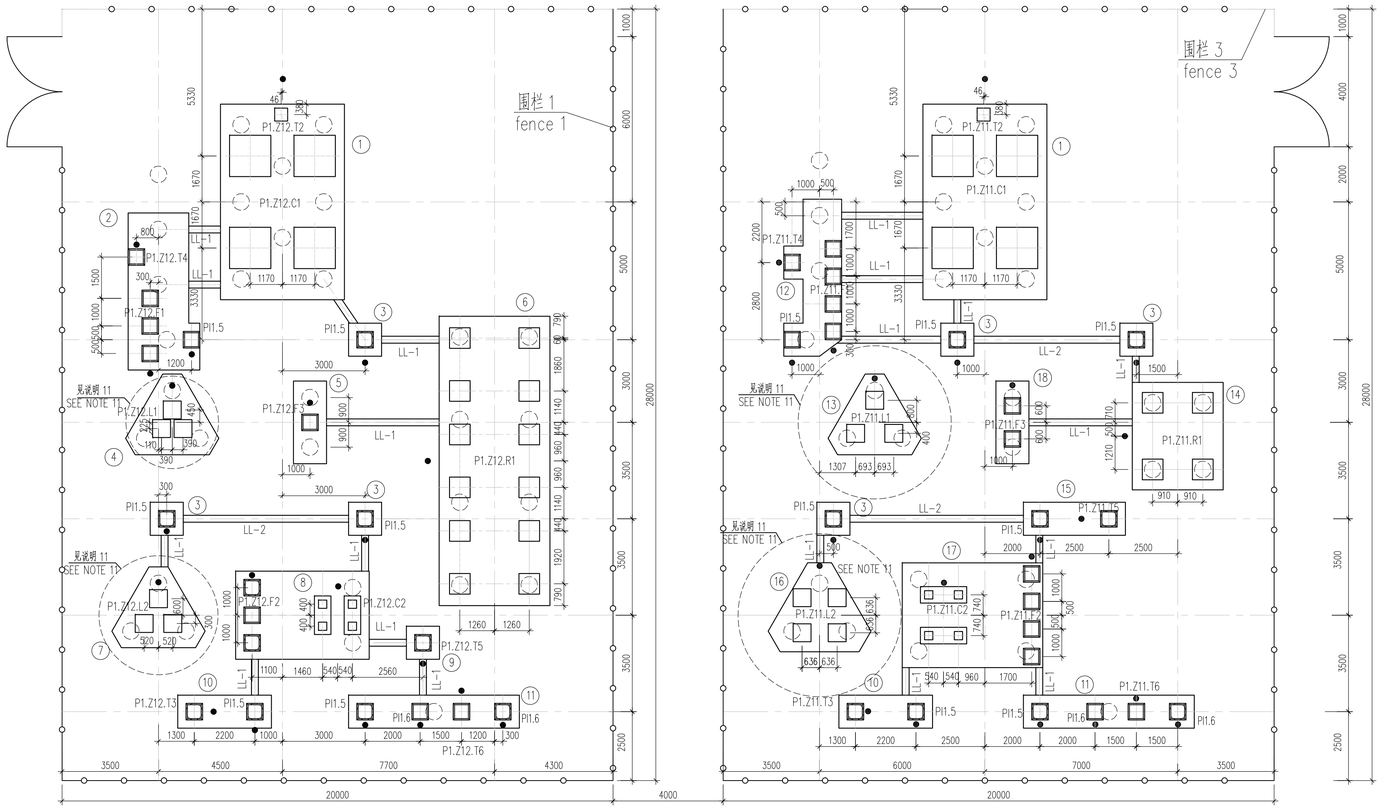
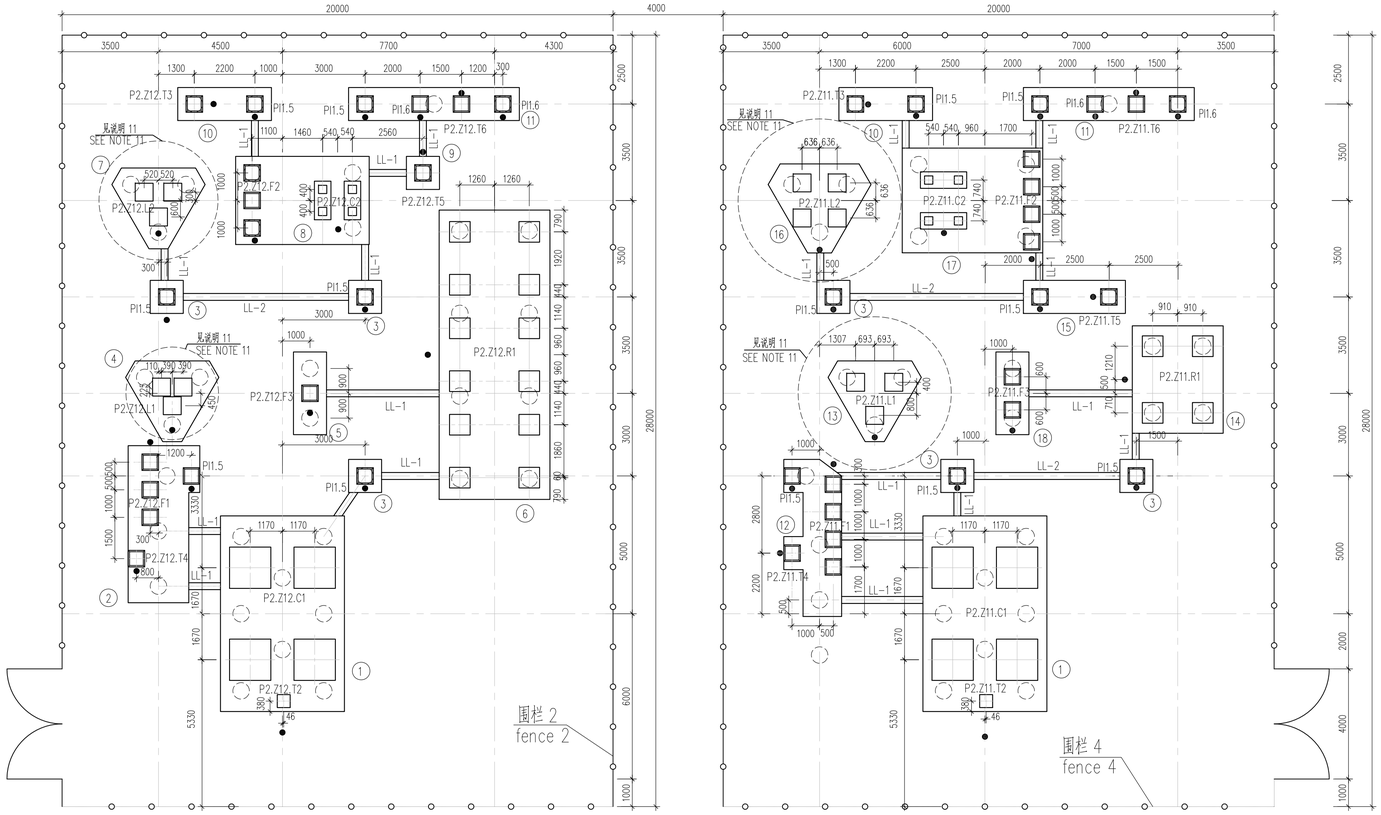
注：The drawings, 1#, 4#, 5#, 6#, are modified.
序号1、4、5、6为本次升版图纸。



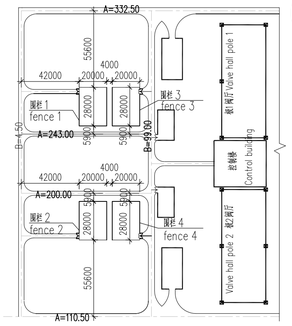


说明:

1、直流场围栏内场地平均标高±0.000相当于绝对标高193.60(EGM2008高程系统)。
如无特别说明，本工程标高均采用相对标高。
2、本工程基础设计依据《巴基斯坦默蒂亚里±660KV直流输电工程拉合尔±660KV换流站施工图设计阶段岩土工程勘察报告（详勘）》(B003701S-G01-01)进行。
3、本工程基础采用柱下独立承台+灌注桩的形式，桩为φ600的机械成孔灌注桩，
4. 材料
1)混凝土
基础强度等级：fc'=28MPa
二次灌浆层强度等级：fc'=35MPa
垫层强度等级：fc'=21MPa
2)钢筋：φ-ASTM A615 Grade 40,Fy=280MPa
Φ-ASTM A615 Grade 60,Fy=420MPa
3)锚栓、螺母、垫片、锚板、定位钢板：Q345B，Fy=345MPa,Fu=470MPa
锚栓、螺母、垫片、锚板均需热浸锌防腐。
5. 混凝土要求：
1)水胶比不大于0.50；
2)最大氯离子含量（水泥用量的百分比）：0.10；测试方法参《ASTM C 1218M》。
3)水泥型号：采用抗硫酸盐的硅酸盐水泥，水泥型号需满足《ACI 318-14》表19.3.2.1对应于暴露等级S1的水泥种类要求(ASTM C150 或ASTM 595 或ASTM C1157对应的类别)；
4)水泥中掺入矿物掺合料(矿物掺合料的掺量、使用方法和耐腐蚀性能按相应产品的使用说明并经验证后确定)。
6. 钢筋保护层：承台底板底部为100mm，其余为50mm。
柱墩和连系梁为50mm.
7. 桩与承台的连接构造参见图集(10SG813)34页 桩顶与承台连接构造，桩内主筋伸入承台长度为35d。
8. 外露基础采用倒圆角处理,R=30mm
9. 本工程初步设计尚未最终审定，最终方案有存在调整可能，是否按图施工由总包CET决定，由此造成的变更设计不予负责。
10. 硬化地坪下方需敷设接地网及埋管，土建需注意施工顺序。
11. 虚线圈中承台及地梁不应合围合钢筋，钢筋绑扎处需采用绝缘材料包裹，见本图钢筋交接处大样。

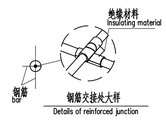
Specification:

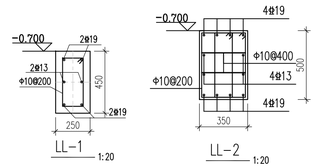
1. The average site level of ±0.000 is equal to the absolute elevation of 193.60 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.
2. The foundation design of the project proceeds according to《The geotechnical engineering investigation report (Detail survey) of drawing construction design phase of ±660KV converter station in Lahore in Pakistan Matiari ±660KV HVDC project》(B003701S-G01-01).
3. The foundation of this project adopts the form of independent cap and filling pile under column. The pile is a bored cast-in-place pile with a diameter of 600.
4. MATERIALS:
1) CONCRETE:
FOR FOUNDATIONS : SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=28MPa
FOR GROUTING: SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=35MPa
FOR concrete cushion: SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=21MPa
2) STEEL REINFORCING:
Φ: ASTM A615 GRADE40, MINIMUM YIELD STRENTH Fy=280MPa
Φ: ASTM A615 GRADE60, MINIMUM YIELD STRENTH Fy=420MPa
3) ANCHOR BOLTS,NUTS AND WASHERS, anchor plate,Locating plate:
Q345B, MINIMUM YIELD STRENTH Fy=345MPa.
MINIMUM TENSILE STRENTH Fu=470MPa.
THE ANCHOR BOLTS, NUTS, WASHERS, anchor plate MUST BE HOT DIP GALVANIZED.
5. Requirements for Concrete:
1) Maximum water-cement ratio is 0.50;
2) Maximum water-soluble chloride ion content is 0.1;Test method refers to "ASTM C 1218M".
3) Type of Concrete: Concrete should be made of sulfate-resistance silica cement. Cement type should fulfill the requirements for concrete used in exposure grade S1, according to《ACI 318-14》table 19.3.2.1.(ASTM C150,ASTM 595 or ASTM C1157 corresponding cement type).
4) Add mineral admixtures (the dosage, using method and anti-corrosion ability should be tested according to the product manual)
6. The minimum concrete cover:
100mm for the bottom of pile cap,50mm for the rest.
50mm for pedestal and the Coupling beam.
7. The connection structure between the pile and the cap refers to the 34 page of atlas (10SG813).The main reinforcement extends into the pile cap with a length of 35d.
8. The surface of pedestal above site level should reverse corner, R=30mm.
9. As the preliminary design documents have not yet been approved finally, the project solution has probability to be modified. The design contractor has no responsibility for the project solution changed, if the construction contractor does not get authorized by CET before construction.
10. The grounding grid and buried pipe are laid under the hardened floor in the fence, and the construction order should be organized.
11. No closed reinforcement should be included in the base and the beam connected in the broken circle, and insulation material shall be used for the binding of the reinforcement, as the detail of reinforced junction in this drawing.



直流场围栏定位图

location of the DC fence





直流场设备支架基础平面布置图（围栏内）

layout of foundations for ±660kV DC equipment support (in the fence)

电气	郑煜	2019.01.28
专业	会签	日期
Speciality	Sign	Date

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 1 3 - 0 0 1 B

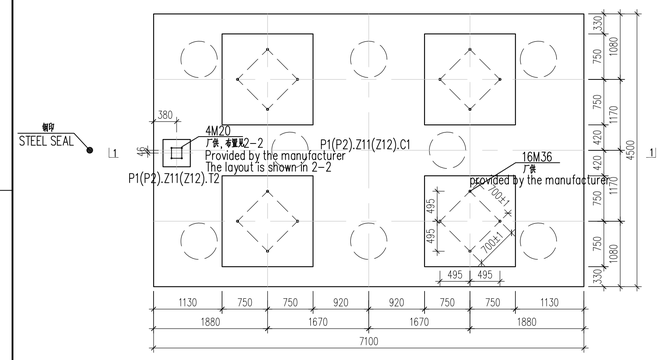
	姓名 Name	日期 Date
设计 Designed	樊树平	2019.01.28
校核 Checked	吉静	2019.01.28
审核 Reviewed	毛清清	2019.01.28
批准 Approved	韩立彦	2019.01.28
比例 Scale	1:100	图号 DWG NO. 30-B003701S-T0313-001

直流场设备支架基础布置图（围栏内）

The layout of foundations for ±660kV DC equipment support (in the fence)

版本 REV. B

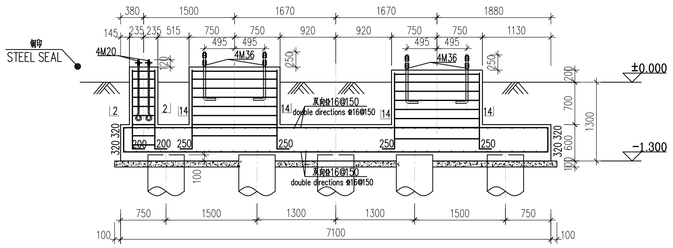
594X841



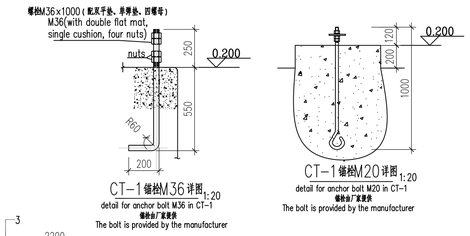
CT-1 1:50

P1(P2).Z11(Z12).C1+P1(P2).Z11(Z12).T2

16M36+4M20



1-1 1:50



CT-1锚栓M36详图 1:20

detail for anchor bolt M36 in CT-1

锚栓由厂家提供

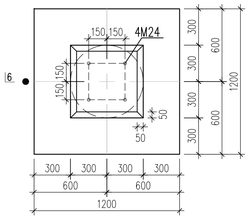
The bolt is provided by the manufacturer

CT-1锚栓M20详图 1:20

detail for anchor bolt M20 in CT-1

锚栓由厂家提供

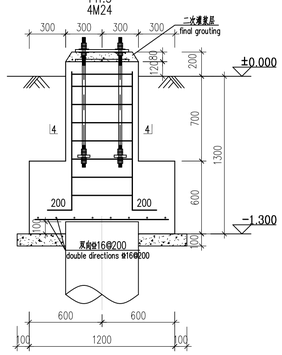
The bolt is provided by the manufacturer



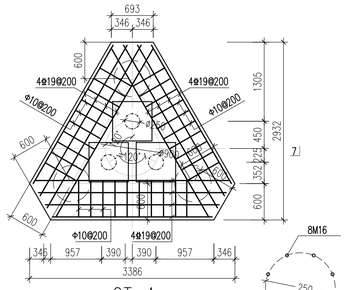
CT-3 1:25

PI1.5

4M24



6-6 1:25

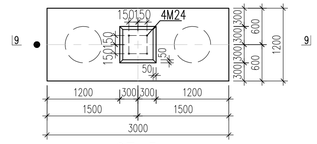


CT-4 1:50

P1(P2).Z12.L1

化学螺栓 厂供，24M16X190，高出基础面60mm

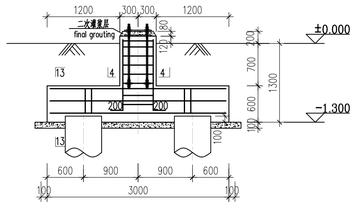
chemical bolt 24M16X190, 60mm above the foundation provided by the manufacturer



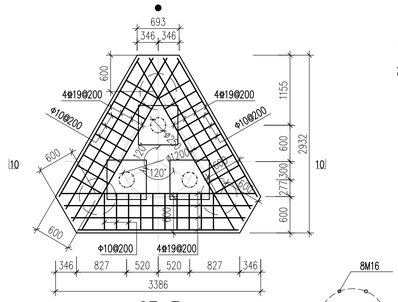
CT-5 1:50

P1(P2).Z12.F3

4M24



9-9 1:50

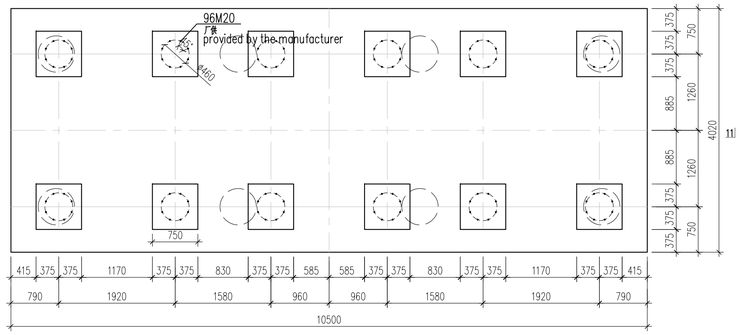


CT-7 1:50

P1(P2).Z12.L2

化学螺栓 厂供，24M16X190，高出基础面60mm

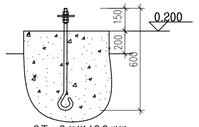
chemical bolt 24M16X190, 60mm above the foundation provided by the manufacturer



CT-6 1:50

P1(P2).Z12.R1

96M20

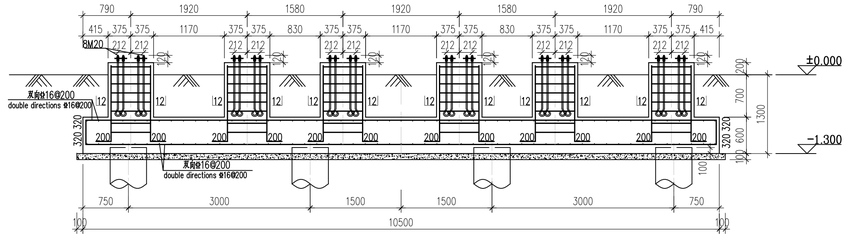


CT-6锚栓M20详图 1:20

detail for anchor bolt M20 in CT-6

锚栓由厂家提供

The bolt is provided by the manufacturer



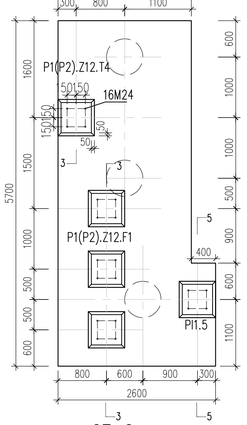
11-11 1:50

说明：

1.● 表示钢印符号。
2.图纸上的尺寸以标注为准，不能按比例量取。
3.地脚螺栓做皮带锁紧的Q345B，需做特殊处理，锁特范围见详图。
4.露出地面基础与地坪间接缝处应设置20mm胀缝，需者混合料填缝。
5.CT-4/CT-7中不应合阀盘，钢筋碍孔处需采用绝缘材料处理，见本图钢筋交接处大样。
6.柱墩间距小于50mm时，可浇筑在一起。
7.其他说明见001图。

NOTE：

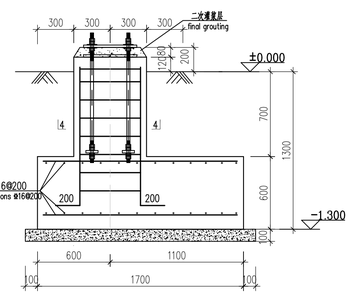
1.● stands for the steel seal.
2.Only written dimensions shall be refered to, and don't measure in scale.
3.The anchor bolts, with Q345B grade, should be galvanized according to the detail in this drawing.
4.The 20mm expansion joint should be set between the foundation and the ground, and the asphalt mixture should be filled in.
5.No closed reinforcement should be included in the foundation CT-4/CT-7, and insulation material shall be used for the binding of the reinforcement, as the detail of reinforced junction in this drawing.
6.if there are columns whose distance is less than 50mm, they can be constructed together.
7.Other notes refer to the drawing 001.



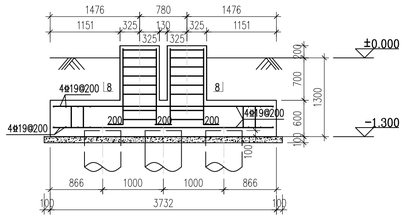
CT-2 1:50

P1(P2).Z12.T4+P1(P2).Z12.F1+PI1.5

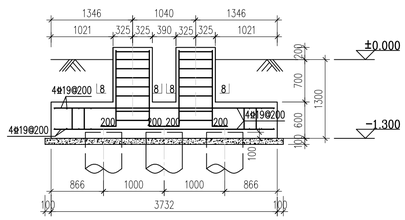
20M24



5-5 1:25



7-7 1:50



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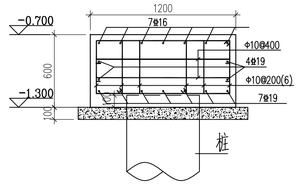
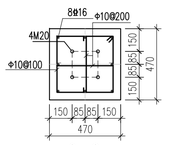
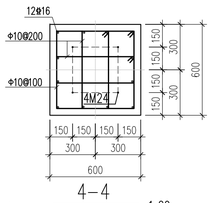


13-13 1:25



2-2 1:20



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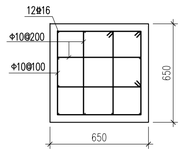

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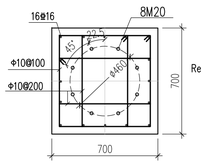
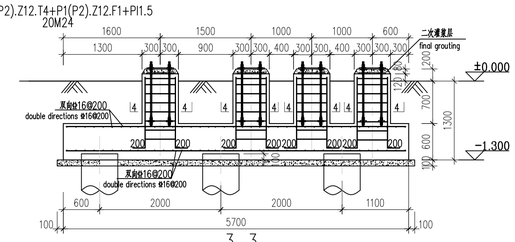
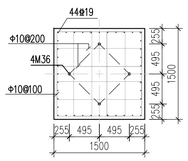


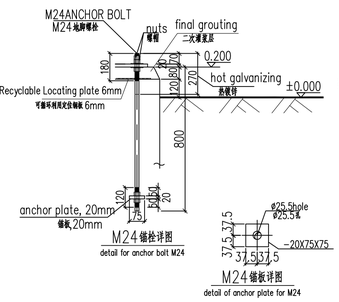
12-12 1:20



3-3 1:50

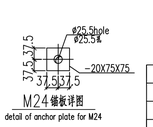


14-14 1:50



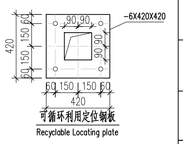
M24锚栓详图

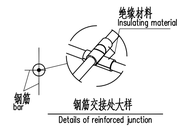
detail for anchor bolt M24



M24锚板详图

detail of anchor plate for M24





钢筋交接处大样

Details of reinforced junction

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project From Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT GROUP CORPORATION

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 1 3 - 0 0 4 B

	姓名 Name	日期 Date
设计 Designed	徐刚平	2019.01.28
校核 Checked	吉韬 陈伟	2019.01.28
审核 Reviewed	邹清	2019.01.28
批准 Approved	韩冬	2019.01.28

CT-1--CT-7详图

Detail of CT-1--CT-7

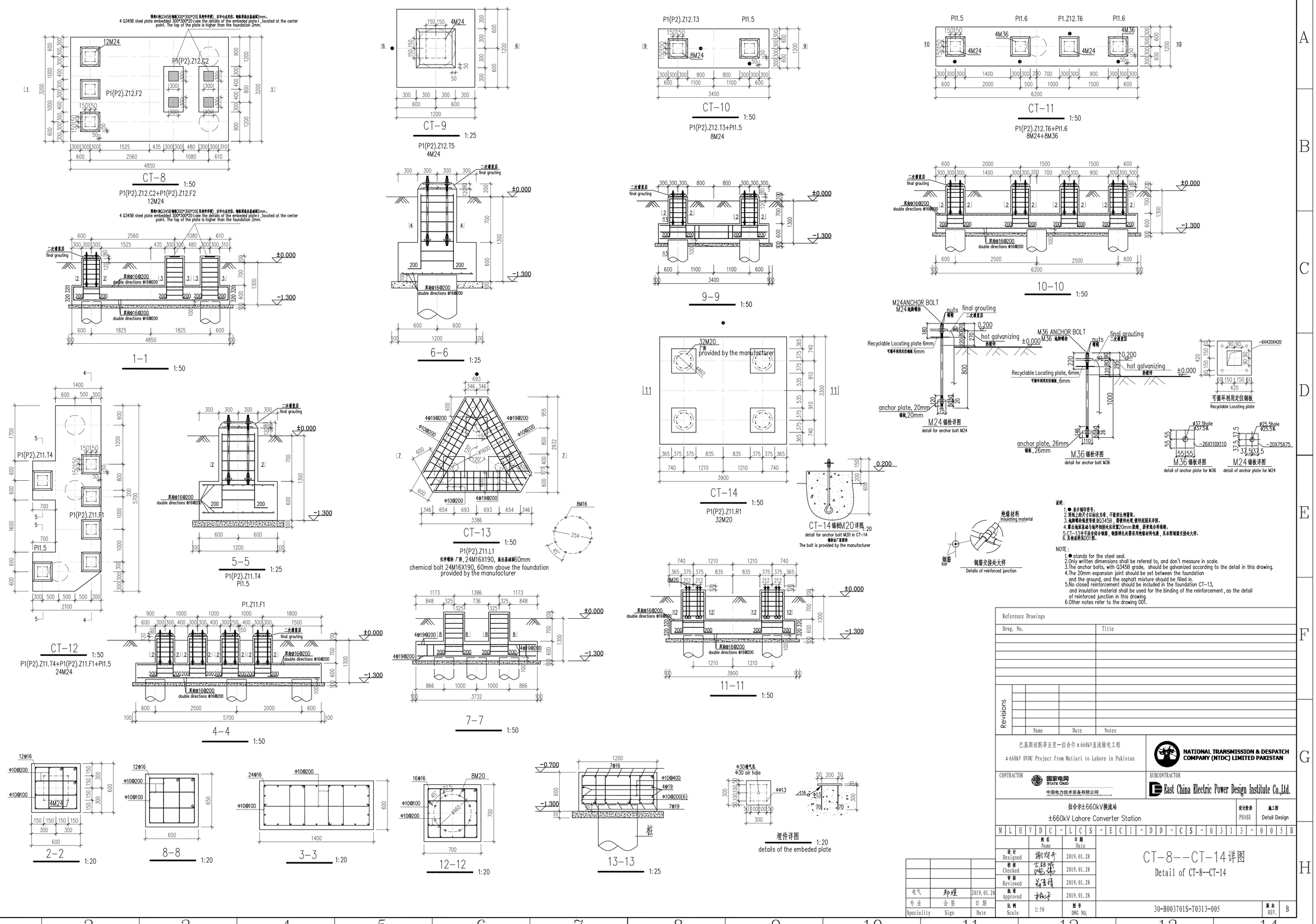
电气	郑煜	2019.01.28
专业 Speciality	会签 Sign	日期 Date

比例 Scale: 1:50

图号 DWG NO.: 30-B003701S-T0313-004

版本 REV.: B

594X841


CT-8 1:50
P1(P2).Z12.C2+P1(P2).Z12.F2
12M24
CT-9 1:25
P1(P2).Z12.T5
4M24
CT-10 1:50
P1(P2).Z12.T3+PI1.5
8M24
CT-11 1:50
P1(P2).Z12.T6+PI1.6
8M24+8M36
1-1 1:50
9-9 1:50
10-10 1:50
6-6 1:25
CT-12 1:50
P1(P2).Z11.T4+P1(P2).Z11.F1+PI1.5
24M24
5-5 1:25
P1(P2).Z11.T4
PI1.5
CT-13 1:50
P1(P2).Z11.L1
化学螺栓 厂供，24M16X190，高出基础面60mm
chemical bolt 24M16X190, 60mm above the foundation provided by the manufacturer
CT-14 1:50
P1(P2).Z11.R1
32M20
CT-14锚栓M20详图 1:20
detail for anchor bolt M20 in CT-14
锚栓由厂家提供
The bolt is provided by the manufacturer
4-4 1:50
7-7 1:50
11-11 1:50
2-2 1:20
8-8 1:20
3-3 1:20
12-12 1:20
13-13 1:25
埋件详图 1:20
details of the embeded plate
M24锚栓详图
detail for anchor bolt M24
M36锚栓详图
detail for anchor bolt M36
M36锚板详图
detail of anchor plate for M36
M24锚板详图
detail of anchor plate for M24
可循环利用定位钢板
Recyclable Locating plate
钢筋交接处大样
Details of reinforced junction
绝缘材料
insulating material
说明：
1. ● 表示钢印符号。
2. 图纸上的尺寸以标注为准，不能按比例量取。
3. 地脚螺栓强度等级为Q345B，需镀锌处理，镀锌范围见详图。
4. 露台地面基础与地坪相接处应设置20mm胀缝，若有混合料填缝。
5. CT-13中不应合闭合钢筋，钢筋绑扎处需采用绝缘材料包裹，见本图钢筋交接处大样。
6. 其他说明见001图。
NOTE:
1. ● stands for the steel seal.
2. Only written dimensions shall be refered to, and don't measure in scale.
3. The anchor bolts, with Q345B grade, should be galvanized according to the detail in this drawing.
4. The 20mm expansion joint should be set between the foundation and the ground, and the asphalt mixture should be filled in.
5. No closed reinforcement should be included in the foundation CT-13, and insulation material shall be used for the binding of the reinforcement, as the detail of reinforced junction in this drawing.
6. Other notes refer to the drawing 001.
Reference Drawings
Drwg. No.
Title
Revisions
Name
Date
Notes
巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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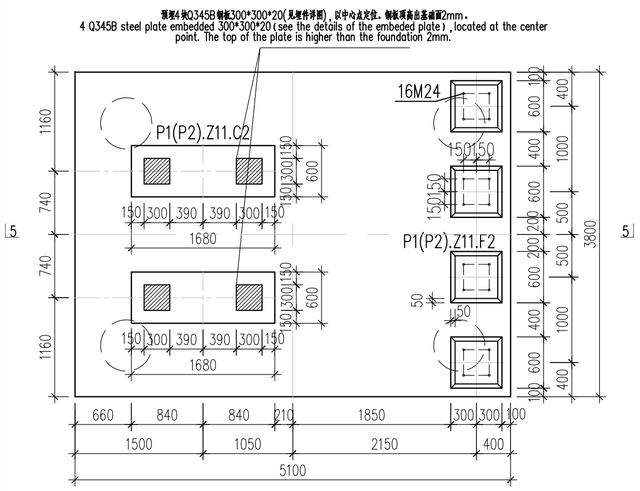
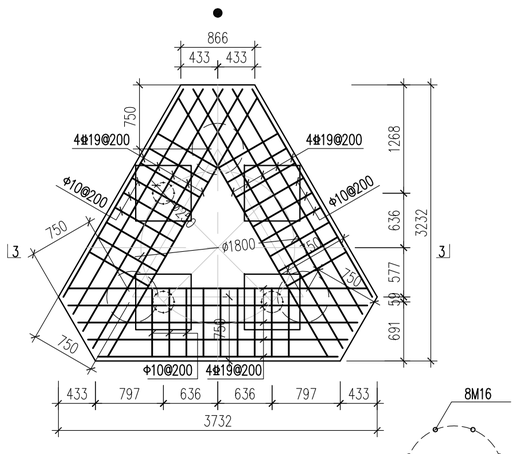
±660kV Lahore Converter Station
设计阶段 PHASE
施工图 Detail Design
M L H V D C - L C S - E C I - D D - C S - 0 3 1 3 - 0 0 5 B
姓名 Name
日期 Date
设计 Designed
2019.01.28
校核 Checked
2019.01.28
审核 Reviewed
2019.01.28
批准 Approved
2019.01.28
比例 Scale
1:50
图号 DWG NO.
30-B003701S-T0313-005
版本 REV.
B
CT-8--CT-14详图
Detail of CT-8--CT-14
电气
郑煜
2019.01.28
专业 Speciality
会签 Sign
日期 Date



CT-17

1:50

P1(P2).Z12.C2+P1(P2).Z11.F2

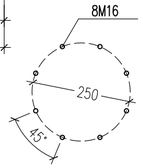


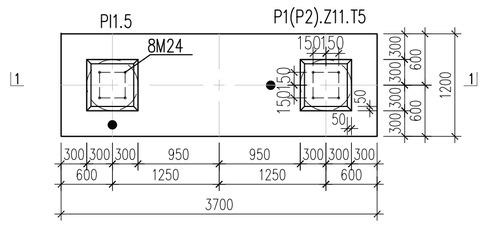
CT-16

1:50

P1(P2).Z11.L2

化学螺栓 厂供，32M16X190，高出基础面60mm
chemical bolt 32M16X190, 60mm above the foundation provided by the manufacturer

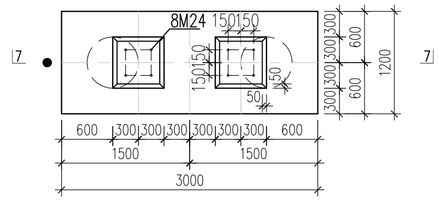




CT-15

1:50

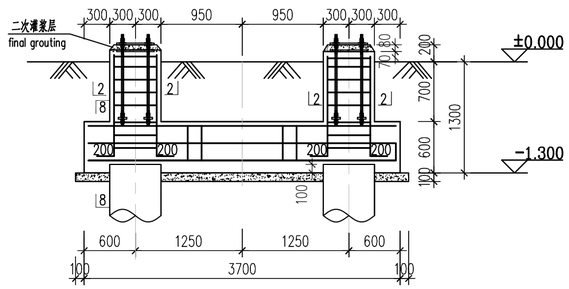
P1(P2).Z11.T5+PI1.5



CT-18

1:50

P1(P2).Z11.F3
4M24



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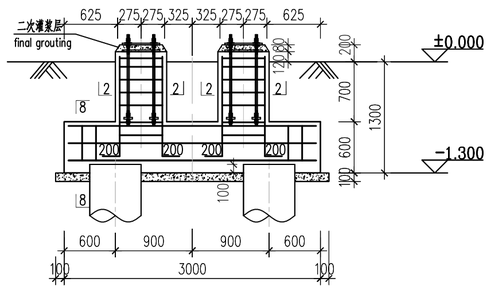
1:50



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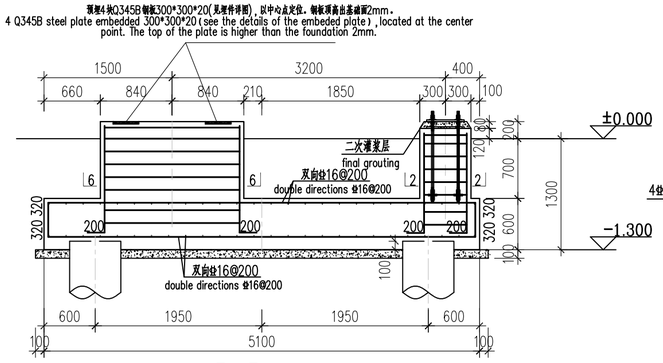

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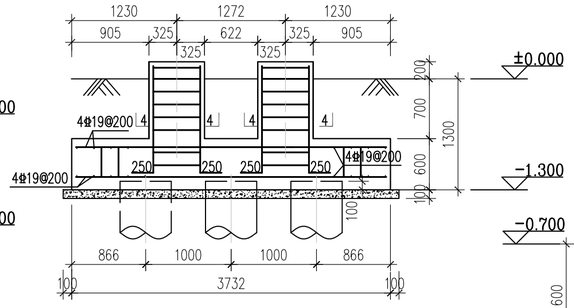

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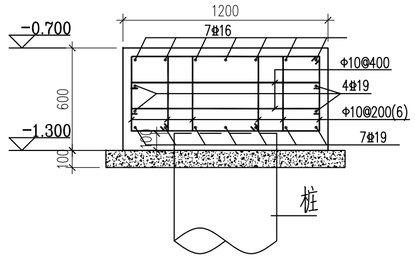

8-8

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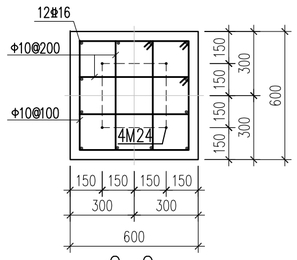
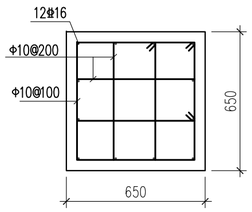


2-2

1:20



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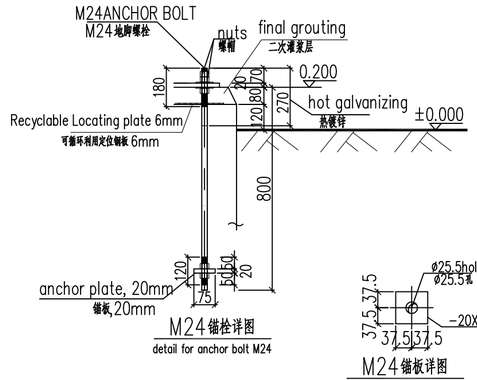
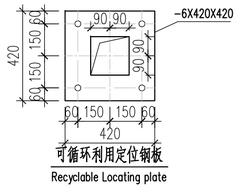
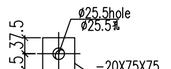


M24锚栓详图

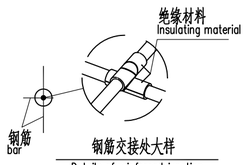
detail for anchor bolt M24





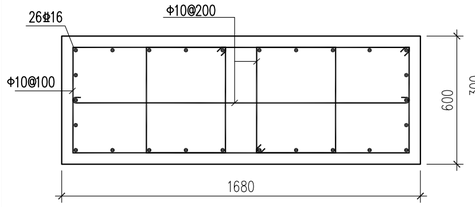
M24锚板详图

detail of anchor plate for M24



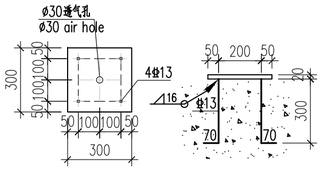
钢筋交接处大样

Details of reinforced junction



6-6

1:20



埋件详图

1:20

details of the embeded plate

说明：

1.● 表示钢印符号。
2.图纸上的尺寸以标注为准，不能按比例量取。
3.地脚螺栓强度等级为Q345B，需镀锌处理，镀锌范围见详图。
4.露出地面基础与地坪相接处应设置20mm胀缝，沥青混合料填缝。
5.CT-16中不应含网合钢筋，钢筋绑扎处需采用绝缘材料包裹，见本图钢筋交接处大样。
6.其他说明见D01图。

NOTE:

1.● stands for the steel seal.
2.Only written dimensions shall be refered to, and don't measure in scale.
3.The anchor bolts, with Q345B grade, should be galvanized according to the detail in this drawing.
4.The 20mm expansion joint should be set between the foundation and the ground, and the asphalt mixture should be filled in.
5.No closed reinforcement should be included in the foundation CT-16, and insulation material shall be used for the binding of the reinforcement, as the detail of reinforced junction in this drawing.
6.Other notes refer to the drawing 001.

电气	郑煜	2019.01.28
专业	会签	日期
Speciality	Sign	Date

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 1 3 - 0 0 6 B

	姓名 Name	日期 Date
设计 Designed	佟财宇	2019.01.28
校核 Checked	古韶卿	2019.01.28
审核 Reviewed	黄清清	2019.01.28
批准 Approved	杨亚	2019.01.28

CT-15--CT-18详图

Detail of CT-15--CT-18

比例 Scale: 1:50 | 图号 DWG NO.: 30-B003701S-T0313-006 | 版本 REV.: B

420X594